

Review Article

Conducting Polymer Nanocomposite for Energy Storage and Energy Harvesting Systems

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Conducting polymers (CPs) have received a lot of attention because of their unique advantages over popular materials, such as universal and tunable electrical conductivity, simple invention approach, high mechanical strength, low weight, low price, and ease of processing. Polymer nanocomposites have been enthusiastically explored as superlative energy generators for low-power-consuming electronic strategies and confirmed progressive surface area, electronic conductivity, and amazing electrochemical behaviour through expanding the opportunity of utilization. The hybridization of conducting polymer with inorganic hybrid and organic nanomaterials also resulted in multifunctional hybrid nanocomposites with better capabilities in a variety of devices, including sensors, energy storage, energy harvesting, and defensive devices. The capability and assistance of modern advancements for the development of multifunctional nanomaterials/nanocomposites have been presented, as well as the approaches for producing nanostructured CPs. The mechanisms underlying their electrical conductivity, and ways for modifying their properties, are investigated. The ongoing research towards generating superior CP-based nanomaterials is also discussed. This assessment focuses on the important schemes involved in the scientific and industrial use of polymeric materials and nanocomposites intended for the scheme and manufacture of energy strategies such as solar cells, rechargeable batteries, supercapacitors, and energy cells, as well as the waiting problems and their prospects.

1. Introduction

Conjugated polymers (CPs) have stayed at the forefront of polymer discipline since the introduction of exceedingly conductive polyacetylenes (PAC) in earlier decades [1]. Inherently, conducting polymers combine charge mobility, redox, and optoelectronic behaviour, which are easier to tune than metals and semiconductors. This results in the mechanical behaviour of organic polymeric entities being hybridized with these more easily tunable properties. Their

range of multifunctional applications has been expanded thanks to the invention of CP nanocomposites and nano-scale morphological control. The full potential of these advantageous characteristics has been realized in actuators, optoelectronics, sensors, energy converting and storing systems, biomedical, and separation technologies or adsorbents for the removal of contaminating entities.

Intrinsically conductive polymers (CPs) are an intriguing class of organic compounds that can be easily synthesized. They exhibit an extensive range of chemical

architectures and a stretchable range of micro- and nano-architectures to achieve tailored chemical, macroscopic, and physical behaviours [2]. Chemically speaking, CPs are conjugated organic polymeric entities with a backbone carbon atomic structure coupled by both bonds and an extended overlap of electron orbitals, resulting in the display of a semiconducting electronic structure [3]. The intrinsic conductivity is caused via doping, which involves injecting or withdrawing electrons from the polymeric chain network's backbone while maintaining perfect electroneutrality by adding opposing ions.

Here, the doping phase is achieved through oxidation, p-doping, or reduction, also known as n-doping reactions. This leads to the development of delocalized charge structural imperfections, such as solitons, bipolarons, and polarons, which are positioned within the energy gap while acting as charge carriers [4]. As a result, these polymeric entities can have their electrical conductivity fine-tuned from insulative to metal ranges via reversible and simple electrochemical or chemical doping/de-doping [4]. Therefore, by simply controlling the doping level, CP electronic structure can be engineered by chemically altering new chemical structures within the molecular range through the application of chemical or electrochemical approaches, leading to the formation of a class of tunable and crucial optically, electronically, and redox/electrochemically attributable properties [5]. As a consequence, an enormous variety of stretchable, lightweight, chemically diverse, and polymeric structures with fascinating properties have been produced for a variety of extremely valuable applications. These polymers, which also include polyaniline (PANI) and its derivative, polythiophene (PT), polypyrrole (PPy), poly(3,4-ethylenedioxythiophene) (PEDOT), polyacetylene (PA) and its derivative, and polyparaphenylene (PPP), have a chain network with a high degree of conjugation in Table 1, while Table 2 displays selected CPs.

However, as a result of advances in nanotechnology, CPs and nanoparticles can now be combined to create polymer nanocomposites (PNCs), bionanocomposites, or nano-biocomposites, depending on the situation [6–8]. A variety of nanomaterials, including 0D (nanoparticles, nanospheres), 1D (nano-ribbons, nanorods, nanowires, nanofibres, nanobelts, and hollow structured nanotubes), 2D, and multisided CPNCs, have been created in the past (nano-rings, nanosheets, nano-disks, nano-clips). Due to their inherent nanoscale nature, CP nanocomposites have undergone extensive examination to determine potential uses [9]. Due to its enhanced electrochemical performance, simple combination, biocompatibility, huge apparent area, high electrical conductivity, raised carrier transport, and distinctive visual characteristics, CPNCs have attracted more interest than traditional polymeric materials.

For the creation of nanocomposites, a variety of physical and chemical methods have been used, such as physical template synthesis (also known as the solid and easy method) and template-free methods (also known as interfacial polymerization, self-assembly, and seeding methods) [10–12].

The acquisitive effects of the aforementioned methods are mirrored in terms of better photo-conversion effect and device constancy. Polymer thin-film-based organic photovoltaic strategies have occurred as a viable different to semiconductor stellar (solar) cells due to their constant flexibility and ascetically cheaper cost. The Buckminsterfullerene derivatives, inorganic nanocrystals, SWCNT and MWCNT, graphene referents, or properly intended donor-acceptor block copolymers are exploited as electron adherent in polymer blended nanocomposites. The unexpected copolymers have been specified to act as donors or acceptors; however, completely conjugated block copolymers, which are prepared of partially isolated but chemically associated donor and adherent blocks, have been recognized as the ablest auxiliary for extremely capable and constant all-organic-based solar cells.

As the human population grows, so does the demand for energy to support quality of life, industrial evolution, and transportation. Natural resources such as petroleum gas, oil, and coal are the primary sources for ecosphere energy supply at the moment. Conversely, vestige fuels have drawbacks such as limited availability, swift depletion, and eco-friendly concerns such as harming air or water [13]. As an outcome, there is a strong desire for low-cost, clean, and renewable energy causes. Noteworthy determinations are being made to create advanced methods for converting solar, wind, or thermal hydropower generation into electricity at a charge that is inexpensive [14, 15]. Organic photovoltaics (PVs), which convert sunlight directly into electricity, are one of these skills that have gotten a lot of attention. The sun's energy has enlightened the earth for three 3×10^{24} J every year, or 10^4 times more than the world's total energy usage. Using only 0.10% of the globe's surface and solar cells through 10% productivity provides enough energy for the entire world [16, 17]. Despite this enormous potential, PV-generated power accounts for only 0.1% of global energy output [18]. Due to the progress of modern light-emitting materials, innovative device design, the advent of applied science or nanotechnology, and revolutions in material blends (in the form of composites or copolymers) or the design of organic-based photovoltaic (OPV) devices, the sunlight to influence power conversion influence (PCE) has prominently improved.

Polymers are the supreme adaptable materials in the world today. To some consecrations over the reverse materials, this is good. Flexibility, tailorability, processability, environmental stability, cost-effectiveness, and lightweight are some of the necessary advantages [19]. Polymers were identified as electrically insulating materials as early as the fifth century. Polymers have been employed as an electrical wire insulating cover, insulating gloves, switches, and a shielding layer on an electronic circuit panel, among other things [19]. Because of the covalent long-chain carbon edge exertion structure, these saturated polymer systems are isolating in nature. For these types of polymers, the molecular orbital bandgap must be more than 10 eV to drive electrons from the valance to the conduction band [20]. Therefore, the insulating polymers usually have a surface electric resistance over $10^{12} \Omega\text{-cm}$. The elasticity of the

TABLE 1: List of some important conducting polymers.

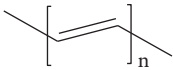
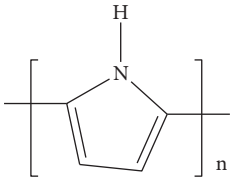
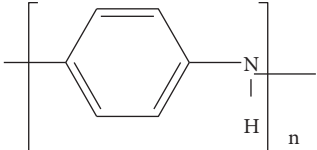
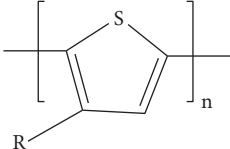
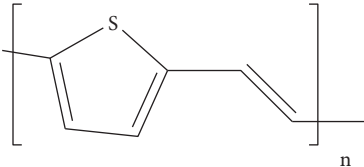
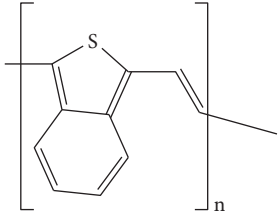
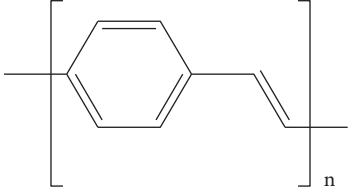
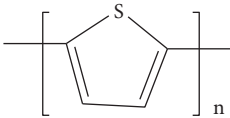
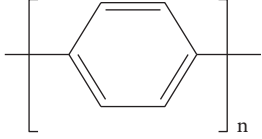
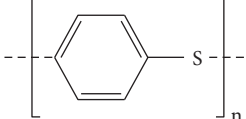
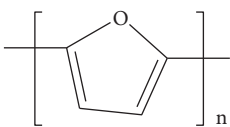
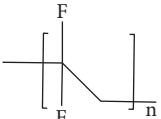
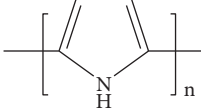
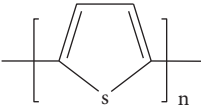
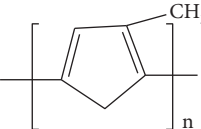
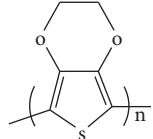
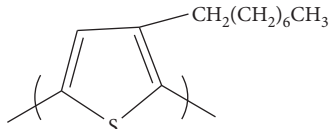
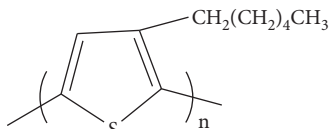
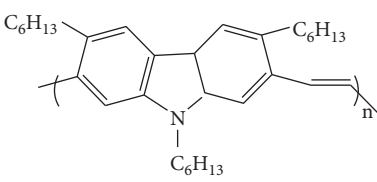
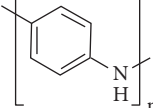
Conducting polymer name	Chemical structure
Polyacetylene	
Polypyrrole	
Polyaniline	
Poly (3-alkythiophene)	
Polythienylenevinylene	
Polyisothionapthene	
Polyphenylenevinylene	
Polythiophene	
Polyphenylene	
Polyphenylene sulfide	
Polyfuran	

TABLE 2: Chemical structure of typical organic thermoelectric polymers.

Polymer name	Chemical structure
Polyvinylidene fluoride	
Polypyrrole	
Polythiophene	
Poly (3-methylthiophene)	
Poly (3, 4-ethylenedioxythiophene)	
Poly (3-octylthiophene)	
Poly (3-hexylthiophene)	
Poly (2, 7-carbazolylenevinylene)	
Polyaniline	

polymer materials has been expanded, as conducting performance is enclosed within the appearance of a numeral of the polymers. In polymers, the physical marvel of conductivity could also be developed generally by two main customs [21]: polymer composite of conducting nanoparticles (carbon nanotube, graphite, metal and inorganic hybrid salt, etc.) with insulating polymers and by π -electron coupling in polymers. The major one is labeled as “ion conducting polymer,” and also, the different one is “intrinsically conducting polymer” (ICP). The models of ion conducting polymers are composites of polyethylene oxide compound [22], polyethylene adipate [23], polyethylene succinate [24] with lithium salts, and so on; individuals of

ICPs are polyacetylene, polyaniline polypyrrole, polythiophene, etc. [25]. ICPs are acknowledged as “synmet” or “synthetic metal” due to the integration of specific metallic appearances, e.g., resistivity and conductivity [26]. The attention to this topic has grown by the day when three scientists (Prof. Alan G. MacDiarmid, Prof. Alan J. Heeger, and Prof. Hideki Shirakawa) were granted the Noble Award for the development and advancement of electrically conducting polymers in the year 2000 [27]. Polymers with conjugated double bonds conduct electricity. The conjugated polymer as π -electrons are lightly bound, and these π -electrons can delocalize throughout the polymer chain [28]. Therefore, there is the bandgap reduction for those

conjugated polymers. Doping, i.e., electron donation to the π conjugated system or electron extraction from the π conjugated system, will increase the conductivity of these polymers by many crinkles. Even if the conductivity of doped conducting polymers has surpassed that of non-conducting polymers, it is still significantly higher than that of conducting metal salts such as Cu, Ag, Au, and Sr, among others (Figure 1). Furthermost of the doped conducting polymer and derivatives define conductivity in semiconductor area (Figure 1). This conducting polymer is extremely much right for potential application as an additional for inorganic semiconductor material.

The present focus in the conjugated polymer sector is on nanoscience and nanotechnology to generate unique, dependable responses for applications such as sensors, energy harvesting, solar cells, and supercapacitor. Conjugated polymer composites containing more variety of nanomaterials are of specific concern because they are associated with the properties of two or more distinct materials, potentially resulting in unique mechanical, electrical, or chemical performance. Therefore, the conducting nanocomposites not only consist of nanomaterials and macromolecules but also address several problems with intrinsic conducting polymers. The well-balanced combining of the qualities of different constituents into a single material can result in novel properties of nanocomposites. Due to its cooperative and hybrid qualities generated from an extensive spectrum of components, nanocomposites of conjugated polymers with nanoparticles have piqued interest. Following nanocomposites have begun to show promise in a variety of sectors, including battery cathodes, microelectronics, nonlinear optics, sensors, energy harvesting, and supercapacitor. Ni et al. [29] recently described the electroactive conducting polymer (ECP) materials as an enormous intimate of carbon-based flexible materials proficient of high-rate storing and provision of electronic energy as of their high electrical conductivity and practicable fast electrochemical kinetic, which can be unique of the perfect electrical energy storage applicant materials to incredulous one of the highly interesting concerns caused by the dynamic confines of the pseudo-capacitive electrodes in high energy density asymmetric supercapacitors. The majority of the ECP constituents undertake a quick redox reaction to produce the pseudo-capacitive reaction, which gives them a higher specific energy than organic double-layer capacitors and a higher level of conductivity than inorganic battery materials, giving them a higher power capacity. The highly conductive polyphenylene backbone of the electroactive conjugated polymers known as polytriphenylamines (PTPAs) appears to be paired with a high energy density electroactive polyaniline. These materials can be produced by a simple natural or electrochemical oxidation of their monomers.

The effective procedure of pseudo-capacitive electroactive conductive polymers (ECPs) for charge storage with assisted electrolyte ion and electron transport is made possible by polymer-based hybrid electrode structures. As a result, the electrode materials have excessive specific capacitance, higher rate ability, and significant driving concert, which are thought to hold excessive potential in an electric vehicle or grid-scale energy storage applications.

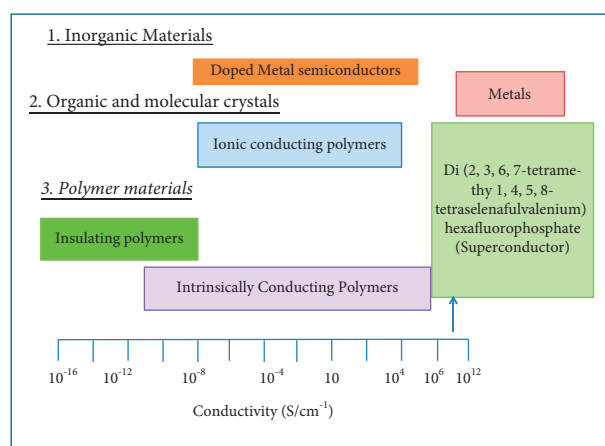


FIGURE 1: Electrical conductivity of materials.

Our main goal is to figure out where the conjugated polymer nanocomposite active in conductometric chemical sensors stands right now. The applications for some conducting polymers are listed in Table 3.

1.1. Types of Conducting Polymers. The outline of π -electron conjugation in a polymer's chain structure can be used to make it naturally conducting. Several covalent bonds are found throughout a π -electron conjugated polymer chain. During the formation of a conjugated polymer chain, all carbon atoms or heteroatoms are sp^2 hybridized. It indicates that one unhybridized p-orbital per atom remains vertical to the polymer chain, while all other p-orbitals persist parallel to one another (Figure 2). As a result, one p-orbital will laterally overlap with either of the closest p-orbitals to form a π -orbital, and the p-orbitals will then be delocalized across the polymer chain (Figure 2). In this instance of isolated covalent bond like ethylene, two π -electrons in nuclear orbital (NO) are localized within the bonding pi-molecular orbital (MO) and antibonding pi-orbital remnants available. Due to delocalization, conjugated polymers containing a double bond, such as butadiene, have two bonding pi-orbitals containing equivalent drives and two antibonding π -orbitals having similar energies relocated to two bonding and two antibonding π -orbitals (Figure 2). As a consequence, the energy of the delocalized maximum filled molecular orbital (HOMO) is enlarged, while the energy of the lower unoccupied molecular orbital (LUMO) is decreased (Figure 3). As an outcome, the energy cavity concerning the occupied bonding orbital and vacant antibonding orbital is decreased. For metal, electrons fill the maximum energy band, while just in the case of semiconductors the highest energy band is partly filled by electrons, whereas in the case of semiconductor polymer HOMO is loaded by electrons, while LUMO stays empty [30]. The necessary energy to transfer an electron from the valance band to the conduction band is required for conduction to occur. The term for this is called "bandgap energy" [30]. The valance electrons are found in the outer shell of electrons in a metal. For conduction to occur, an electron must gain

TABLE 3: Applications for some of the most imperative conducting polymers in energy harvesting.

Polymer	Possible applications
Polycetylene	Solar cells, rechargeable battery, organic photovoltaics, chemical and gas sensors, radiation detectors, Schottky diode, antielectrostatic, encapsulation, biotechnology, optoelectronics, etc.
Polypyrrole	Schottky diode, rechargeable battery, condenser, printed electronic circuit boards, chemical sensors, electroplating, electroacoustic device, fillers, adhesive, transparent coating, electromagnetic shielding, electro-photochemical cells, field effect transistor, photocatalysts, physiological implantations, optoelectronics, conductive textiles, and yarn
Polythiophene	Optoelectronics, rechargeable battery, display device, fillers, field effect transistor, Schottky diode, gas and chemical sensor, photocatalysts, and other devices
Polyaniline	Chemical and biosensor, rechargeable battery, electrochromic devices, indicator devices, textiles, etc.
Poly (p-phenylene)	Photocatalysts, rechargeable battery, fillers

enough energy to be raised to the conduction band, where it can easily transfer. The prohibition gap is thought to be the energy difference concerning the valance and conduction bands. In case of metal due to the “electron sea model” structure, the valance band edges with the conduction band (Figure 3). As a result, metals such as Cu, Au, Ag, Fe, and others are useful conductors. The difference between the two bands in an insulator, such as rubber, Bakelite, or wood, is large, and electron progress to the conduction band is inaccessible. The constrained conductivity is found in materials with a moderate forbidden bandgap. Semiconductive materials, such as GaAs, ZnO, conjugated polymers, and others, fall into this category. Between conductors and insulators, they have transitional conductivity. External energy or defects enable them advanced the conduction of electricity. The polythiazyl (SN)_x, the first conjugated inorganic polymer, was found in 1975. It shows functional conductivity and develops great conductive at 0–29 K [31]. Shirakawa et al. [32], who discovered that the inorganic iodine (I) doped trans-polyacetylene, (CH)_x, with a conductivity of 10³ S·cm^{−1}, established the presumed organic polymer as an electrically conducting one in 1977. Since that time, there has been a growing interest in synthesizing various organic conjugated polymers with conducting properties. Polyaniline (PANI) and its derivatives, polythiophene (PTh), polypyrrole (PPy), polyfuran (PFu), poly(p-phenylene), and polycarbazole are examples of other conducting polymers [32, 33]. Table 1 shows the necessary conducting polymers.

1.2. Conduction Mechanism in Conducting Polymers. The supposed “electron sea” concept of the metal can accurately describe the metal’s high conductivity. The conduction band’s lightly certain mobile electrons are responsible for the metal’s functional conductivity. Those electrons will simply be portable across the atoms in the lattice [34]. The conductivity (σ), which is common to the electric resistivity (ρ) for metallic, can be determined with the help of realistic Ohm’s law:

$$V = IR,$$

$$R = \frac{\rho l}{A}, \quad (1)$$

where V , I , R , l , and A are voltage, current, resistance, probe distance, and area separately. Ohm’s law is fully obeyed by the conducting metals that are upgraded to the status of resistance unit materials. However, this law does not apply to all materials, such as one-dimensional conductors (linear polymer chains), gas discharges, vacuum tubes, and common semiconductors. The conductivity of those components is determined by the number density (number of electrons; n) and the speed with which electrons flow through the materials (charge mobility; μ). So, the conductivity of these materials can be measured by the following equation:

$$\sigma = n\mu e, \quad (2)$$

where e is the electron charge. Because of positive charge carriers, semiconductors and electrolyte solutions must be added to the term (holes and cations). The aberration of resistance unit behaviour for orthodox semiconductors and conjugated polymers could also be described from the assumed conduction mechanism. In case of inorganic semiconductors, the conduction band is unfilled and there is some energy gap between the unfilled conduction band and also the filled valance band. As a result, some external force (doping) is required to generate conducting materials by narrowing the bandgap by adding or deducting electrons from the structure. This added electron (electron donation) or hole (electron withdrawal) can move around the inorganic semiconductor’s lattice, resulting in conductivity. Conduction mechanisms in conjugated polymers differ significantly from those in conventional inorganic semiconductors. The doped conjugated polymers have charged types such as soliton, polaron, and bipolaron [35–38]. In case of conducting conjugated polymers, the conductivity depends on the concentration of the charged ions. For conduction in conjugated polymers, another essential parameter is “carrier mobility,” that is, how fast the charge will move. The conjugated polymers have three commands for the carrier tractability exclusive to the polymer system [34]:

- Single chain or unit passage
- Interchain or intermolecular transport (hopping)
- Interparticle transport (percolation)

Thus, the mobility and, therefore, the conductivity of the conjugated polymers are determined on separately microscopic (inter- and intramolecular) and macroscopic

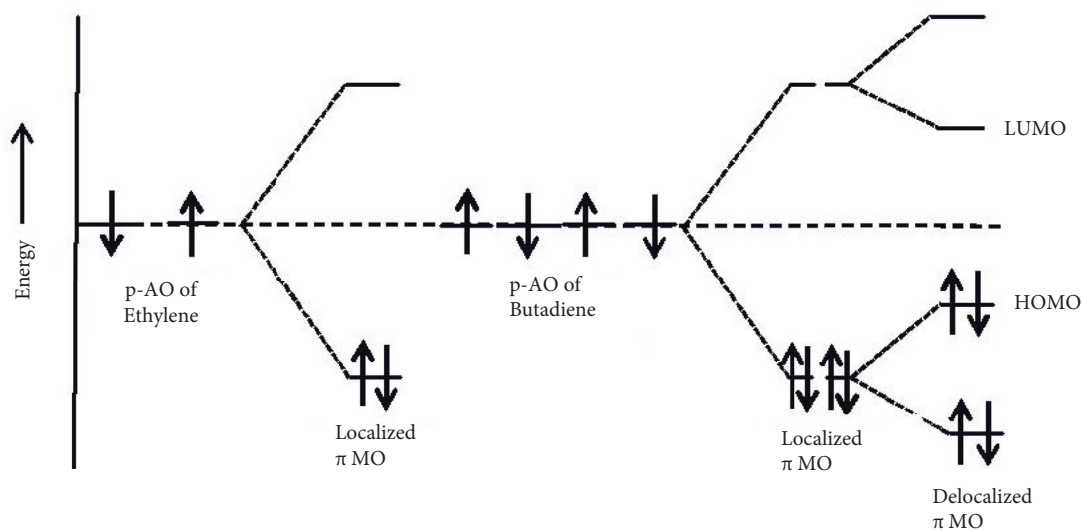


FIGURE 2: Illustration of π -MO energies for localized and delocalized systems.

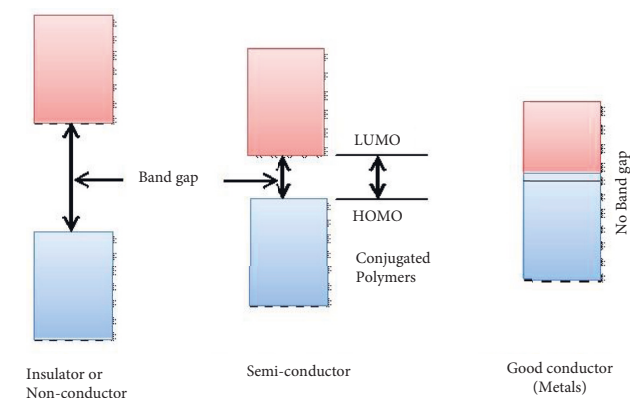


FIGURE 3: Band gap constructions for insulator, semiconductor, and conductor.

(interparticle) level. The above three forms of mobility combined with a sophisticated resistive network govern the effective mobility of the control carriers. The challenge, on the further, is tracing the charge carrier's journey through the polymer.

As this can be model state of concern once the polymer is the same ordered that means crystalline. However, all of the conjugated polymers are semi-crystalline in nature; i.e., they comprise amorphous disordered domain with some crystalline ordered region. Alternative determination is essential to cause the transport of charge between the advanced boundaries established by the multiple numbers of phases [34].

1.3. Polymer-Based Energy Harvesting System. The numerous electronic devices created on conducting polymers have been presented in recent years. These electroactive materials are used in a wide range of requests, from solid-state technology to biotechnology and energy harvesting. Feasible applications of a number of the essential

conducting polymers are recorded in Table 2. The stability in the atmosphere, quantity, density, and deposition is the most important parameters for selecting those materials. Organic polymers, which were acknowledged as electrical, insulating applications (e.g., low dielectric coatings, switches, insulating cover on wire, and insulating electrical apparatus), have emerged as an innovative class of category of electronic materials. Conducting polymers with an extensive range of electrical conductivity (10^{-5} – 10^3 S·cm $^{-1}$) have exciting visual and mechanical features in addition to metallic conductivity. As a result, these conducting polymer structures could be employed to replace inorganic electrical, optoelectronic, and semiconducting resources. The leads of using conducting polymers instead of inorganic constituents include design freedom, tailorability, versatility, low weight, and environmental stability. Conducting polymers with electrochemical, electrical, and optical activity can be used as parts of antistatic additives, electromagnetic screens, corrosion inhibitor coatings, organic light-emitting diodes, micro-electro-mechanical actuators, electrochromic mirrors, and ultracapacitors or supercapacitors [39]. Conducting polymers has also piqued the interest in the research and industrial communities due to their potential applications in a variety of disciplines.

1.4. Polymer as Light Harvesting Material. The photovoltaic properties of solar cells are measured using power adaptation potency, which is calculated by separating the determined harvest power by incident power [40]:

$$\eta = \frac{P_{max}}{P_{in} J_{SC} \times V_{OC} \times FF / P_{in}}. \quad (3)$$

Here, J_{SC} is short-circuit current density; V_{OC} is open-circuit voltage (V); and FF, fill factor, is defined by the consequent equation, which is revealing of the connection to quadrilateral of the form of J–V curves.

$$FF = \frac{J_{P_{max}} \times V_{P_{max}}}{J_{SC} \times V_{OC}}. \quad (4)$$

The short-circuit current density, open-circuit voltage (V), and fill factor should all be enhanced to advance photovoltaic concert. The bandgap of light harvesting materials is usually closely related to the short-circuit current density. Photons with energy created in the bandgap can be captured when a solar cell is lit, creating excitons. The external quantum efficiency (EQE) of a light harvesting system is defined as the ratio between the calm photo-generated charges and the range of incident photons. The short-circuit current density could be determined by the following equation:

$$J_{SC} = e \int_{E_g}^{\infty} EQE(E) \times nAM1.5(E) dE, \quad (5)$$

where E_g is the bandgap of light absorber, and $nAM1.5(E)$ is number of photons with entirely different energy in the normal incident spectrums. The minor bandgap is capable of additional light absorption thus higher JSC.

The alternative aspect that influences solar cell performance is open-circuit voltage, which is determined by the bandgap of organic photoactive materials. The energy modification between the lowest unoccupied molecular orbital (LUMO) of the acceptor and the highest occupied molecular orbital (HOMO) of the donor determines the open-circuit voltage in polymer solar cells [41]. The functional open-circuit voltage is by trial and error represented in the following equation.

$$V_{OC} = \frac{1}{e} \left(|HOMO_{Donor}| - |LUMO_{Acceptor}| \right) - 0.3V. \quad (6)$$

Besides, the LUMO of the acceptor should find at a lower place than the LUMO of the donor, as a result of electrons being transferred from the donor to the acceptor. When working with photoactive materials, it is important to strike a balance between the narrowed bandgap and the increased HOMO/LUMO energy differential to sustain a high open-circuit voltage. The fill aspect, which is related to the microscale shape of the active layer, charge removal layer, and posttreatment of the confession method [42], is the third component that influences solar cell performance. The fill factor has an effect on the short-circuit current density and open-circuit voltage (V) of a device when compared to internal series and comparable resistance.

Novelties in materials science and technology have resulted in promising behaviour for achieving high photovoltaic performance. In bulk heterojunction polymer solar cells, the shape is especially important due to the inherent properties of photoactive polymer materials. As a result, a keen empathy of three critical topics is required to build extremely efficient polymer solar cells: material design, morphology control, and interface engineering.

1.5. How and Why Polymer Nanocomposite-Based Energy Harvesting System? Several conducting polymer and related variants are commonly generated by two primary

approaches, organic or electrochemical oxidation of appropriate monomers. The conducting polymers are thiophene and indole, furan, carbazole, azulene, aniline, etc. Other products are the essential monomers' assistance in the assembly of developed conducting polymers.

Figure 4 depicts the electrical conductivity and vital possible uses of conducting polymer nanocomposite based on carbon nanotubes, inorganic nanomaterials, and conductive polymers, depending on the inherent conductivity of various nanomaterials. Compared with carbon-based nanoparticles and conductive polymers, metal-based nanostructures have greater conductivities [43]. Due to their affluence in constructing conductive networks within elastic matrices, 1D-shaped nanomaterials such as carbon nanotubes and nanofibres with high aspect ratios have the highest conductivity when compared to 0D and 2D nanoparticles and nanosheets. Flexible conductor applications advantage from stretchable nanocomposites with higher conductivity, while strain sensor applications benefit from those with comparatively minor conductivities.

Two-dimensional (2D) carbonaceous nanomaterials beyond graphene have opportunities, problems, and perspectives in the field of energy storing, which have been covered in [44]. The 2D materials, which include carbonaceous nanosheets and graphene, layered transition-inorganic metal chalcogenides, and MXenes, are a class of innovative materials with special properties because of their small dimensions, enormous visible surfaces, distinct facets, continuous conducting pathways, open abbreviated diffusion distances, simple strain relaxation, and prolonged interlayer spacing for advanced energy conversion and storage. These inexpensive 2D carbons have a great deal of potential as graphene substitutes and could potentially be employed as controllable size conducting networks to support active components.

Chemical oxidative coupling polymerization is used to make the bulk of redox polymers, and it is a very easy and accessible process. The aqueous or carbon-based protonic acid result, the monomer, and oxidant are simply grouped. Ammonium persulfate ($(NH_4)_2S_2O_8$), ferric chloride, H_2O_2 , potassium dichromate ($K_2Cr_2O_7$), $CeSO_4$, and other oxidants are commonly utilized. The conducting polymer has electrochemically put on the effective electrode prepared of materials such as Pt, iron, Au, indium tin oxide, metal oxide, glass, and other materials during electrochemical synthesis. The infusibility and excellence of ICPs in common solvents are generally a key difficulty when it comes to incorporating external components in ordinary mixing ways to make nanocomposite. As a result, synthesis processes have to be developed and perfected to encapsulate the nanoparticles in the conducting polymer, giving the nanocomposite research a distributed dimension. The successful mixing of the qualities of the parent constituent's channel some interface can yield novel properties of nanocomposite. The size of the distributed nanoparticles, as well as inequalities within the polymer matrix, influences the properties of the nanocomposite. Those facts can be managed or changed during the nanocomposite synthesis.

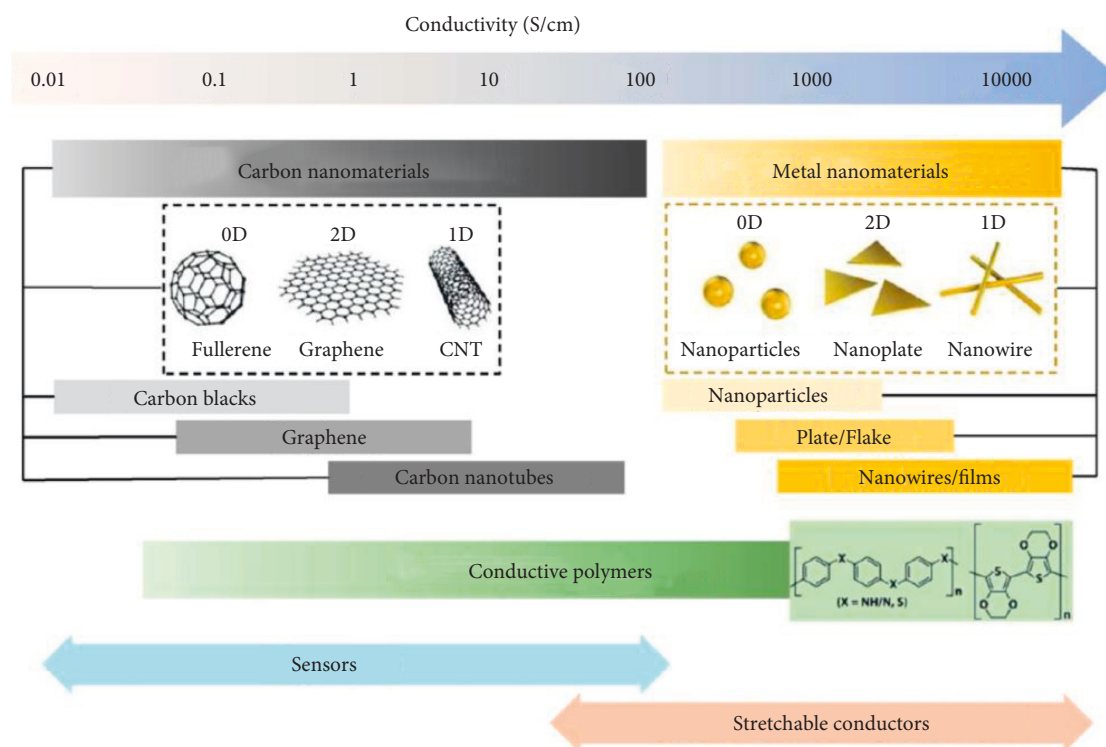


FIGURE 4: Conducting polymer with different carbonaceous nanomaterials and their advanced applications [43].

The procedures *in situ* and *ex situ* using as organic or electrochemical processes have been used to successfully create hybrid inherently conducting polymer nanocomposites in the past. The *in situ* process for preparing conducting polymer nanocomposite has been improved due to differences in nanomaterial dispersity and, as a result, the effect on nanocomposite properties.

Certain nanoparticles are varied within a continuous matrix comparable to conjugated polymer in polymer nanocomposite. The nanomaterials are the materials in which at least one of the material's measurements is of the nanometre directive. The transition from micro to nanoscale materials causes substantial variations in physical characteristics because of their large surface area for a given size. This could be because surface area and surface characteristics influence various necessary chemical and physical relationships. Figure 5 depicts reciprocated nanomaterial shapes and their corresponding surface area-to-volume ratios. In general, these materials are divided into three categories based on their geometries: element, covered, and stringy nanomaterials [45]. The carbon black, nanosilica, metal and metal oxide nanoparticles, and polyhedral oligomeric silsesquioxanes can be categorized as nanoparticle reinforcing causes, whereas nanofibres and carbon nanotubes (SWCNT and MWCNT) are examples of fibrous materials. The nanofiller is defined as a covered nanomaterial once it has a nanometre width and a high percentage of layer-by-layer structure (such as an organosilicate). Advanced nanoparticles are being used to create nanocomposites including conducting polymers. Active variations in composite qualities could also be generated based on the request of the

components utilized and the preparation technique. Several surveys on the occurrence of nanoparticle incorporation techniques into polymeric matrix have been reported. The dimensions of scale as phases, as well as the degree of combination between the two phases, have a considerable impact on the properties of a nanocomposite. To realize the full potential of nanoparticles' technological uses, it is necessary to have some interaction between them, resulting in an innovative of hybrid materials identified as "polymeric nanocomposites." The nanocomposites could also be derived by combining conjugated polymers with nonconductive nanomaterials such as nanoclay or the conducting and semiconducting nanomaterials such as metal, metal oxide, and carbon nanotube (CNT).

1.6. Ex Situ Methodology. An *ex situ* approach is defined as one in which nanomaterials are manufactured or combined within a pre-synthesized conducting polymer as matrix (Figure 6). The ICPs will be fitted first in the *ex situ* approach by oxidative polymerization as organic or electrochemical of a consistent monomer. Over time, chemical or electrochemical methods will be used to integrate nanoparticles into the polymer matrix. Chemical processes yield powdery or dusty nanomaterials, which can be upgraded to polymer nanocomposite through a chemically pre-synthesized conjugated conducting polymer using an *ex situ* and *in situ* approach during the polymer's synthesis. Nanoparticles can also be cycled in the framework of a cluster polymer by intermingling a suitable solution phase, which is the most often utilized method. Alternatively, the

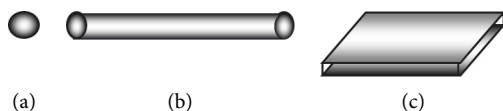


FIGURE 5: Forms of common nanomaterials. (a) Nanolayer. (b) Nanorod or nanofibre. (c) Nanoparticle.

various nanoparticles can be chemically modified in the conjugated polymer film casting process by solution desorption. For example, during the film casting of the derivative of conducting polymer PANI as poly(*m*-aminophenol), a single-step technique for the manufacture of silver (Ag) nanoparticles is created by simple thermal decay of AgNO_3 ammonia complex [46]. The electrochemical separation is also an embedded technique for creating conjugated polymer nanocomposite films. The electrochemical ex situ approach, for example, has been proven to be effective for incorporating metal nanoparticles into pre-deposited polymers. The ex situ electrochemical blend of conjugated polymers and metal nanocomposites is a two-step procedure. By electro-oxidation of a suitable monomer, an ICP that appreciates polyaniline, polypyrrole, or poly(methyl thiophene) is sited on a probe in the first step. Before electrochemical reduction, polymer sheets are dipped in a reaction comprising metal salts of Pt^{6+} , silver (Ag^+), or copper (Cu^{2+}), giving Au, Pt, Ag, Al, or Cu nanoclusters enclosed in the conducting polymer [47]. In this result, metal nanoparticles storing the outward, for example, is normally the case when inorganic metal particles are put onto pre-deposited polymers by metal ion reduction [48]. Ex situ systems allow for the precise and consistent distribution of inorganic nanostructures inside the polymeric matrix, as well as the possibility of fine-tuning the ordering ability. To the fictitious nanoparticles, the nanocomposite in the proper polymer host systems is widely disseminated. Ex situ approaches, on the other hand, allow for the direct transfer of inorganic nanomaterial's principal size-dependent properties into the mass matrix and their successful clustering with well-defined polymer features. In the ex situ development, due to the high outward energy of nanoscale ingredients, the integration customs reinforced the application of an usual solvent often explains the existence of share isolation or aggregation, which causes destructive failing of the thermal, mechanical, electrical, and optical features of the critical nanocomposite material.

1.7. In Situ Methodology. In this procedure, the nanomaterials can be diverse with monomers earlier the polymerization within the polymerization medium and the monomer is polymerized to make the nanocomposite (Figure 7). In the in situ technique, pre-produced nanomaterials are incorporated into the polymerization standard before polymerization, or nanomaterials are directly synthesized in the polymerization standard through polymerization from the consistent pioneer. Because the nanoparticles used are worried by the monomer dynamic positions and before polymerized, the resulting

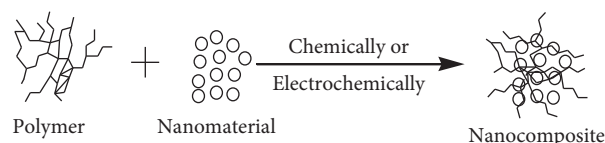


FIGURE 6: Ex situ nanocomposite synthesis method.

nanocomposite has enhanced dispersibility. The in situ approach can be chemical or electrochemical in this case as well. The polymerization of the specific monomer is over in the occurrence of premade nanomaterial colloids at little monomer, and oxidant attention was significant to synthesizing two constant colloids in the chemical route. By electro-polymerizing, the monomer in the existence of pre-synthesized nanoparticles as a colloidal diffusion and nanomaterials can be electrochemically delivered into a conjugated polymer matrix. Several techniques have been devised to chemically attribute pre-synthesized nanoparticles to a conjugated polymer film through the electro-deposition method. The nanocomposite can be synthesized in a variety of ways, including covalently attaching metal nanoparticles to the conjugated polymer mainstay [49]. During electrochemical polymerization or copolymerization with various derivatives, gold (Au) nanoparticles are mixed with polythiophene derivatives. Recently, oppression oligothiophene halted nanoparticles [50] and announced a system for uniformly distributing metal (Au) nanoparticles in a polythiophene matrix. The most difficult aspect of the in situ method is controlling the preparative and development circumstances, which intensely affect the properties of the final nanocomposite material, as well as the relatively weak connections concerning the designed nanomaterial and the host matrix, which limits the polymer's ability to coordinate the nanocomposite area. To this moment, a variety of inorganic hybrid and metal oxide nanoparticles have been surrounded within the structure of intrinsically conducting polymers, resulting in a set of polymer nanocomposite.

1.8. Effect of Nanomaterial Interaction on Conjugated Polymer Appearances. The conducting polymer nanocomposite proposals are a simple technique to mix the structures of conjugated polymer materials with those of inorganic nanomaterial. The active interfaces of inorganic nanomaterials with polymer matrix, which are more useful to reinforce the qualities of conducting polymer composites with nanomaterials than those of primeval conjugated polymers, are largely dependent on the features of conjugated polymer composites with nanomaterials. Nanocomposite production aims to have the conjugated polymer in solution/dispersion while also educating the thermal, mechanical, and electrical properties for practical uses. The effective, long-lasting, or crucial interaction of nanomaterials with the polymer matrix provides the basis for their uniform dispersion. The interface with the monomer during polymerization may synergistically boost the nanoparticle's dispersibility. For example, electrochemical conducts where aniline is active to solubilize SWNTs by the

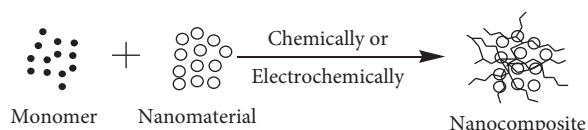


FIGURE 7: In situ nanocomposite synthesis technique.

development of donor-acceptor pioneering are used to build a polyaniline/SWCNT nanocomposite film with improved electroactivity and conductivity [51]. The conductivity of a doped conjugated polymer composite containing nonconducting nanomaterials is usually lesser than that of the doped conjugated polymer. As a result, the goal of nanocomposite research is to improve thermal stability, mechanical strength, flame resistance, and molecular barrier qualities for anticorrosion coating demands, rather than to increase conductivity. The refined anticorrosion coatings are nanocomposites of conjugated polymers, such as polyaniline, polyaniline derivatives, poly(3-alkylthiophene), and polypyrrole, with improved or rare multilayered sodium montmorillonite (Na-MMT) clay. Two sodium ion (Na^+) and calcium (Ca^{+2}) salt tetrahedral sheets are sandwiched between two edge-shared octahedral sheets of either magnesium (Mg) or metallic element hydroxide in the natural chemical structure of MMT. The crucial interaction of conjugated polymer with hydrophilic pure clay in nanocomposite has been demonstrated. The doping interface for the doped conducting polymer can be uniformly interrupted by this type of noninteracting nanoparticle. For example, when nanoclay or nanofillers are applied to dodecyl benzene sulfonic acid ($\text{C}_{18}\text{H}_{30}\text{O}_3\text{S}$) doped polypyrrole, the conductivity drops dramatically because of effective doping interruption [52]. Usually, changes in a variety of physical properties of conducting polymers piqued people's interest. Particularly shaped particles were created, which improved the polymer's compactness and the ordering of composite particles. In recent years, low amounts of additional nanomaterial templates have sparked a lot of interest in developing nanostructured polymer composites with the nanomaterial and thus proposing a route for desirable qualities through the enhanced nanostructured imperative of a preferred morphology. Silica-polyaniline-core-shell polymer nanocomposite, for example, has been created, in which silicon dioxide (SiO_2) nuclei serve as templates for aniline monomer adsorption as thriving as pledge ions for doping of the generated polyaniline (PANI) [53]. The electrochemical polymerization approach was also used to create nanocomposites for conducting polypyrrole, polyaniline, polyaniline its derivative, and polythiophene derivative using Al_2O_3 nanoporous templates [54]. With the help of pyrrole, gold, and inorganic metallic element, nanoparticles were electrodeposited [55]. The connection between conducting polymer and various nanomaterial by electron donor and adherent is suitable for applications in electronic or nano-electronic methods because it can dope the polymer. For example, some of the exterior charges taken by Fe_3O_4 nanoparticles are transmitted to the polyaniline in their nanocomposite by Fe_3O_4 nanoparticles, which operate as an added dopant to increase the conductivity of the

polymer nanocomposite. The nanoparticles' particular ion ability can act as dopant counter ions for the conjugated polymers in nanocomposite systems, such as V^{5+} ions in V_2O_5 -polyaniline system. Conducting polymers with carbon nanomaterial (CNT) composites have recently attracted a lot of attention due to their ability to advance the thermal, morphological, electronic, and mechanical properties of polymers, which has opened up innovative potentials for chemical sensors and energy harvesting. Owing to charge transfer doping and site perceptive interface with the conducting polymer constraint, nanocomposites of conjugated polymer with multiwalled carbon nanotube (MWCNT) have attracted a lot of attention. It has the potential to produce polymer-based composites in which the carbon nanotubes are not just mixed in with the polymer, but are in close touch with it through particular charge transfer interactions. The conductivity of nanocomposites has been enhanced by charge transfer interfaces between conjugated polymer and π -conjugated surfaces of carbon nanotube (CNT) [56, 57]. SWCNT and MWCNT/polyaniline composite films, for example, were created using an electrochemical system in which the donor-adherent complex, *i.e.*, week redox doping effects, boosted electrochemical activity and composite conductivity. The carboxylic acid group functionalized multiwalled carbon nanotube (MWCNT) for nanocomposite production may increase this type of modest doping impact in conjugated polymer/MWCNT nanocomposites. As an example, when the $[-\text{COOH}]$ group functionalized MWCNT (c-MWCNT) was exposed to an active dopant, the conductivity of the polyaniline/c-MWCNT nanocomposites increased. In addition, the carboxylic acid group functions as a dopant in carbon black core nanocomposites with conjugated polymer [48]. The polymerizing polyaniline (PANi) nano-pillars group onto hierarchical 3D consistent porous N-doped carbon nanofibre (HCNF) scrolls; rolled-up fibre tubes with outer directional porous and internal hierarchical porous structures have been created as electrodes of fibre-shaped supercapacitors (FSCs) (PANi-HCNFs) [58]. The rolling electrospun PAN nanofibres film up into a microfibre produced the HCNF scrolls. When associated with pure carbon nanofibres (CNFs), HCNF scrolls could offer a better specific surface and a better associated porosity network. The inner and outer hierarchical porous structures, as well as the high N-doping concentration, may assist with effective electrolyte transport to the inner area of the composite fibre and with increased electrical conductivity, which will help with improved electrochemical performance. The 3D PANi-HCNF scroll electrode as-built thus demonstrated extremely flexible capacitance concert in terms of ultrahigh specific capacitance of $339.30 \text{ F}\cdot\text{g}^{-1}$ ($85.10 \text{ mF}\cdot\text{cm}^{-1}$), high energy density of $11.60 \text{ Wh}\cdot\text{kg}^{-1}$ ($3.00 \text{ Wh}\cdot\text{cm}^{-1}$), and good cycling strength with retention of 74.20% after 3000 cycles at $0.5 \text{ A}\cdot\text{g}^{-1}$.

1.9. Organic-Inorganic Composites/Hybrids. Inorganic semiconductors' configurable energy phases, perfect bandgap, high mobility, and moral stability have fuelled

their use as photoactive energy harvesters. The strong optical immersion, configurable bandgap/vitality levels, simplistic reaction processability, and large hole mobility of conjugated polymers are combined with the high electron movement, highly electroactive, good thermal, mechanical, and scope customizable optoelectronic features of NCRs to create these hybrid devices. While blending with soluble polymers, solution-processable NCRs provide a high interfacial region for efficient exciton dissociation. When organic and inorganic processes are combined in a heterojunction device, the conducting polymers operate as donors, attracting solar light and transporting hole, while the NCRs act as adherents, allowing electrons to flow. Indeed, it has been demonstrated that many inorganic NCRs, when used in conjunction with conducting polymers, act as sunny gathering acceptors with functional transport things, allowing them to produce their carriers, dissociate excitons created in conjugated polymers, and efficiently passage carriers to the assembling electrode [59]. The power conversion efficiency of further than 3% has been achieved using similar procedures [60].

Inorganic NCRs, such as cadmium selenide CdS, PbSe, PbS, CdTe, zinc oxide, SnO, titanium oxide, and Si, have previously been shown to be active in inorganic hybrid solar cells. Because the outline and scope of NCRs affect authentic energy levels and bandgap, efforts have been prepared to modify the NCRs. By controlling the growth conditions, cadmium selenide of various sizes (e.g., diameters and lengths) and forms (e.g., nanoparticles, nanorods, and tetrapods) may be created [61–63]. The chemistry synthesis methods can also produce TiO₂ nanoparticles, nanorods [64, 65], and carbon nanotubes [66, 67], as well as ZnO nanorods [68, 69] and nanowires [70–72].

Liu et al. [73] via solution-based association of close-packing monolayers of block copolymer (BCP) micelles, including 2D electrochemically EG, graphene oxide (GO), MoS₂, and TiO₂ nanosheets and 1D CNTs, confirmed a reliable and adaptable protocol for controlled patterning of mesoporous conducting polymers of polypyrrole (PPy) and polyaniline (PANI) on various functional free-standing surfaces. With this patterning method, a number of 2D ultrathin hybrid nanomaterials with distinctive sandwich structures, controllable pore diameters, and variable thickness can be successfully produced. When used as supercapacitor and micro-supercapacitor (MSC) electrodes, this innovative structure improves electrochemical capacitance and rate performance.

Furthermore, this flexible approach would offer a new direction towards the controlled synthesis of 2D hybrid materials with well-defined mesoporous architectures and potential applications in a variety of fields given the variety of offered BCP templates and strong interactions with functional surfaces and precursors (such as inorganic hybrid metal, metal oxides, and organic polymer) [73].

To obtain their charge separation capabilities and organic photovoltaic ability, these NCRs are integrated into different conducting polymer-based host matrices [59]. To examine the charge transfer from conjugate polymer to CdSe using photoluminescence (PL) extent, cadmium selenide has

been combined with various conducting polymers containing CN-PPV, MEH-PPV, and MEH-CN-PPV [74]. To find the optimal combination, CdSe NCRs with radically diverse element sizes were also put into conducting polymer matrices. In a single-junction hybrid solar cell, the basic design criteria for donor polymers are to drop their bandgap to a maximum of 1.5–1.6 eV with a LUMO level, that is, E_b greater than the inorganic acceptor CB. The polymer then absorbs a broad range of sunlight while maintaining its V_{oc} . The band chart of typical conjugated polymers and NCRs with visibly defined energy levels (i.e., HOMO and LUMO sites) could be used as a strategy reference to select the correct CP/NCR mixture for an organic photovoltaic technology. This could be used as a strategy reference to visualize the achievable while choosing the right CP/NCR mixture for an organic photovoltaic technology. The inorganic NCRs as electron acceptor should have a larger EA than donor CPs for heterojunction solar cell applications. The conduction band of acceptor inorganic semiconductors should be lower than LUMO of donor polymers in a variety of situations, making it potentially beneficial for exciton division and charge relocation at interfaces. Furthermore, in solar cell applications, strong electron-accepting ability and electron mobility are necessary.

As can be observed, the conduction band of numerous NCRs is lower than the LUMO of several conducting polymers (CPs), allowing these NCRs to act as electron acceptors for those specific conjugated polymers, allowing exciton separation and charge allocation at the CP/NCR heterojunction to occur. It is worth noting that the energy levels come from different patron and acceptor materials; yet, when they are mixed to form interfaces, these energy levels can fluctuate due to contact dipoles and further phenomena. As a result, caution must be exercised when designing with the authentic device, to account for several discrepancies in the energy state caused by balances at the edges.

The ordered heterojunction (OHJ) and the polymer-inorganic BHJ are two common architectures for organizing hybrid solar cells [71]. The polymer-inorganic hybrid BHJ solar cells, such as polymer-fullerene BHJ solar cells, would address the confines of bilayer devices with limited patron-acceptor interfacial space and poor exciton separation. To improve solution processing, NCRs could be surface altered to make them decipherable in organic solvents (such as CHCl₃, solvent, and C₆H₅Cl).

To modify the surface characteristics of NCRs, a variety of surface comforting ligands were applied. The conjugated polymer/NCR systems can be created by combining the two elements physically to make hybrids/bulk nanocomposite or by in situ synthesis of one equivalent into the other to form proper nanocomposites (NCs). During a polymer matrix, TiO₂ will be dissolved in the polymer solution to create hybrids with a diffusing system. In addition, appropriate CP monomers with discrete NCRs can be polymerized in situ to create organic/inorganic NCs. Since they offer direct charge transport channels and well-controlled heterojunctions, OHJ solar cells are commonly observed as the most capable form for nano-hybrid solar cells. Inorganic semiconductors,

such as nanotubes, nanofibre, nanorods, and nanowires, can be vertically attached to substrates, forming nanoporous design, which can then be infiltrated with conjugated polymers to detect the OHJ molecular structure. However, due to inadequate polymer filling induced by incoming CP particles delaying the nano-pores, this technology for producing ordered heterojunction hybrid solar cells is extremely inspiring. The pore size, relative molecular weight of the polymer, and solvent used to blend the conjugated polymer and moderating circumstances all influence polymer filling in nano-pores [75].

Innovative techniques such as in situ producing the conjugated polymers in the pores by UV light-assisted polymerization [76], chemical oxidative approach [77], or electrochemical polymerization method [78, 79] have been described to explain the challenges with considerably filling polymers into nanostructured pores. The TiO_2 nanotubes were lordotic in 2,5-diiodothiophene solution before being irradiated with ultraviolet light in an argon atmosphere, resulting in the separation of (C-I) bonds produced from reaction products and the creation of polythiophene, which was coupled with the TiO_2 nanotube outward. The device was designed to compare the device performance of polythiophene penetrated into TiO_2 nanotubes to that of in situ polymerized polythiophene on TiO_2 nanotubes. UV irradiated in situ polymerized polythiophene on TiO_2 showed a thousand times higher current density than penetrated polythiophene inside TiO_2 , implying efficient exciton separation due to enhanced coupling concerning polythiophene and TiO_2 nanotube.

By in situ synthesis of NCRs, conjugated polymers can also be chemically associated with inorganic nanocrystals for enhanced charge transfer from polymer to inorganic acceptors. By combining diethyl zinc precursor with P3HT solution, where it experiences hydrolysis, continued by condensation response and annealing the film at 100°C , zinc oxide (ZnO) NRCs were in situ produced inside the P3HT polymer medium [80]. With the help of the in situ produced ZnO NCRs within the P3HT matrix, system strength of 2% was attained at 520 nm wavelength, with an EQE of 44%.

1.10. Superfluous Applications of Polymer Nanocomposites.

Conducting polymer nanocomposites (CPNCs) have improved electrolytic diffusion in batteries and supercapacitors, improved ferroelectric polymeric-oriented capacitor dielectric behaviour, facilitated capable separation in solar cells, reduced response time, improved sensitivity in biosensors and chemical sensors, and improved corrosion inhibition [81–83]. Therefore, this study provides a thorough explanation of recent developments in conductive polymer (CP) development, as well as multi-size nano-architecture and conducting polymer nanocomposite.

We provide insight into the various methods used to synthesize conducting polymer nanocomposites (CPNCs). Additionally, CP composites' electrical conductivity characteristics and applications are given. Additionally, as schematically depicted in Figure 8, the representative conducting polymer nanocomposite research studies for

specialized devices are also highlighted [84]. The potential difficulties in science and technology for synthetic methodologies and applications are also clarified. As a result, this study goes into detail on market structure, conducting polymer nanocomposites and conducting polymer bionanocomposite (CPB) manufacture, characterization, characteristics, and emerging applications.

1.11. Supercapacitors. Supercapacitor gadgets made of conducting polymers are considered in three categories: type {I} (symmetric) utilizing indistinguishable p-dopable polymer for every electrode, type {II} (asymmetric) utilizing two disparate p-dopable polymers of a restrictive kind of electroactivity, and type {III} (symmetric) utilizes equal polymer for the two terminals of p-doped structure as the positive electrodes and the n-doped structure as the negative electrodes [85]. The conducting polymers' potential during this application, notwithstanding, has not been taken advantage of yet, because of certain restrictions of conjugated polymers, most altogether connected with their somewhat low controlling dependability and restricted versatility of anions in common conjugated polymer layers. The chemical hybridization and composites along with various inorganic/natural materials intend to conquer the impediments. To benefit from the features of conjugated polymer-based supercapacitor, scientists had tried to blend conjugated polymer-based double composites, for example, conjugated polymer-metal oxides, conjugated polymer-carbon materials, and the conjugated polymer-based ternary composites, such as composite consisting of conjugated polymer-carbon-metal oxide as supercapacitor channel dynamic materials to increment electrochemical execution of supercapacitor through harmonious impacts.

Flexible electrodes, supercapacitors, batteries, and sensors are just a few of the devices that use complexed graphene nanocomposites, along with other polymers, and inorganic hybrid nanomaterials [86] (Figure 9). The layer-by-layer approach works by sequentially adsorbing components with opposing charges via striking forces such as electrostatic interfaces and hydrogen bonds [87–93]. To manage the thickness and configuration of hybrid composite materials at the nanoscale, extremely ordered and multi-layered designs created using layer-by-layer association can be produced in a consistent manner.

Wearable electronics increasingly depend on high energy density, high-voltage stretchable fibre-shaped supercapacitors (SFSSs). Wang et al. [94] showed off a brand-new fibre electrode with exceptional qualities such as high electronic conductivity ($1771.77\text{ S}\cdot\text{cm}^{-1}$) and exceptional electrochemical characteristics. With the help of this fibre, researchers were able to create an innovative flexible fibre-shaped supercapacitor with an extensive electrochemical window (1.62 V), high areal and volumetric energy densities ($8.3\text{ Wh}/\text{cm}^2$ and $6.6\text{ mWh}/\text{cm}^3$, respectively), high areal and volumetric power densities ($400\text{ W}\cdot\text{cm}^{-2}$ and $320\text{ mW}/\text{cm}^3$, respectively), and constant cycling performance (89% capacitance retaining after 5000 cycles). The electrically connecting numerous flexible fibre-shaped

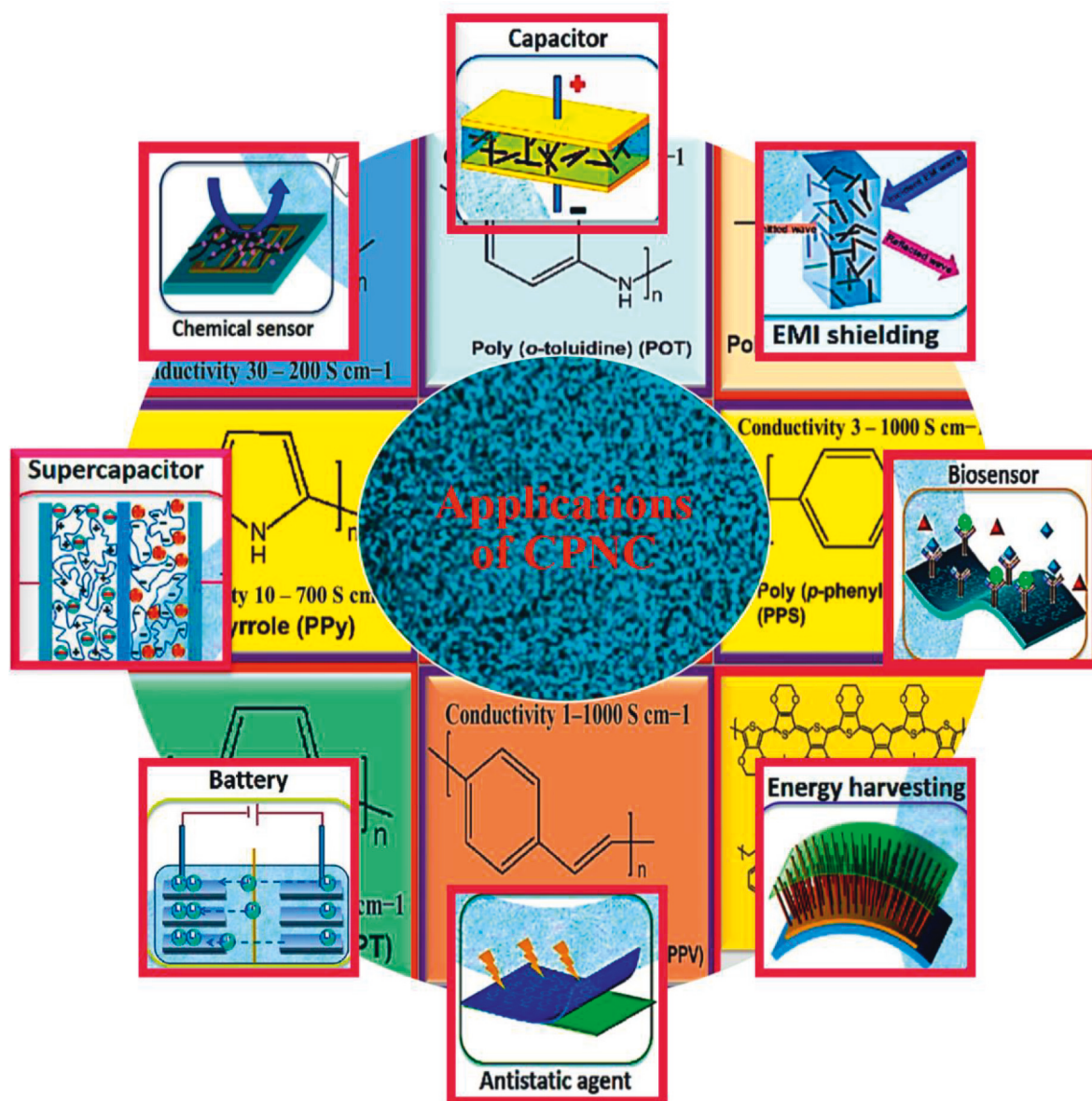


FIGURE 8: Illumination of different conjugated conductive polymer nanocomposites in innovative applications [84].

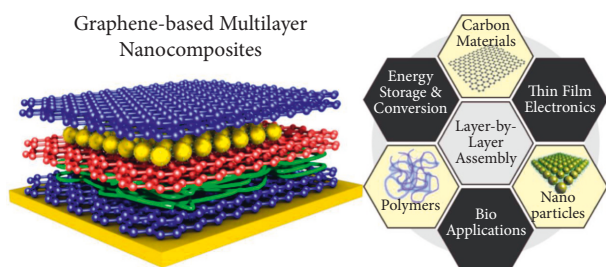


FIGURE 9: Carbonaceous nanomaterials with polymers using layer-by-layer method in energy harvesting applications [86].

supercapacitors (SFSSs) in series without utilizing metal wire interconnects, and these fibres' excellent conductivity also made it possible to make stretchable cycle supercapacitors

quickly and easily. The resulting cycle groups (T-SFSS) comprised of 8 serially connected cells demonstrated high-voltage output of 12.8 V, extremely high energy density of $41.1 \mu\text{Wh cm}^{-2}$ at power density of $3520 \mu\text{W}\cdot\text{cm}^{-2}$, and unusual stretchability of up to 400% without explicit capacitance decline. This work offers a new family of flexible fibre electrodes and innovative theory designs for flexible power systems that might be incorporated or stitched into wearable and convenient electronics.

In the vast majority of realistic models, each part can give with their characteristic electroactivity anyway the actual connection comes from various parts of the mixture arrangement. Outstandingly, on account of appropriately planned composites, the definite capacitance of the fusion is not unprejudiced by a straight blend of the constituents; however, in ideal conditions, the total volume of the mixture

will be greater than the sum of its parts. The composites ordinarily get minor charge transfer resistance associated with the pure conducting polymer that fosters the exhibition in electrochemical capacitors. The ternary composite as ready by Cho et al. [95] (inorganic metal oxide nanoparticles/PEDOT: PSS/graphene) showed a large electrical conductivity of $1570 \text{ S}\cdot\text{cm}^{-1}$ and large energy thickness of 73 Wh/Kg along with a regular stock of 81.5% after 1,000 phases. Xia and collaborators [96] developed a nano-composite electrode with primary shell plan upon graphite foam (graphite foam + Co_3O_4 /PEDOT- MnO_2 core/shell nanowire arrays). The metal oxide and polymer with graphite foam showed the most elevated energy thickness of 9.8 Wh/kg , and the electrode achieved 20000 cycles with 90.2% holding limit.

In addition, hybrid supercapacitor devices lie in which the conducting polymer's positive electrode is shared with the negative-activated carbon electrode. Composite electrode materials, on the other hand, are being considered for addressing the needs of technological applications in a variety of ways. In both acidic and neutral electrolytes, a major shell PPy/PANI composite conveyed through in situ chemical oxidative polymerization of aniline monomers upon the surface of polypyrrole (PPy) nanotube conditional excellent electrochemical achievement. The PPy/PANI composite confirmed a definitive capacitance of 416 F/g in $1 \text{ M H}_2\text{SO}_4$ electrolyte and 291 F/g in 1 M KCl electrolyte [97]. The PPy-based completely solid-state redox supercapacitor with a PMMA-centred gel electrolyte had been explored to solve the scientific complications of oxidation and escape associated with the observation of liquid electrolyte. Cells showed the capacitive behaviour of the greater amount of capacitance $15.3\text{--}22.5 \text{ mF/cm}^2$ (equivalent to diverse electrode-specific capacitance of $120\text{--}178 \text{ F/g}$ of polypyrrole (PPy) and regularly small amount of interior resistance because of the elastic character of the liquid electrolytes. The capacitive performance of the redox cell is similarly contrary to the kind and extent of the cations of the salts utilized in the gel electrolytes, according to a simulated investigation [98]. The exhaustive analysis of distorted (ITO/steel) and symmetric (ITO/ITO and steel/steel) supercapacitor procedures composed of nano-metric PEDOT films revealed that the asymmetric ITO/steel arrangement has the highest definite capacitance and notable electrochemical stability, similar to that found for progressive PEDOT-inorganic hybrid nanocomposites. The ultraporous formation of the ultrathin films placed over ITO, which control continuous audience oxidation-reduction processes as well as the distinct electroactivities of the two electrodes, is credited with the asymmetric supercapacitors' unique characteristics.

The use of graphene subunit integrated polymeric material or change intermediates among innovative polymer and graphene for energy storage tasks was demonstrated by Li et al. [99]. When utilized as an electrochemical material in batteries, sensor, and supercapacitors, the graphene polymers have a significant number of graphene subunits and different pores that allow for rapid and efficient electron and ion transport networks. Chemical efficient groups on

organic divisions of these graphene polymers or nanopolymer enhance the electrochemical achievement of supercapacitors by incorporating pseudo-capacitive chemical groups for supercapacitor uses. If the molar ratio of tetra propyl ammonium-manganese oxide (TPA-MO) and PPy progressed to 1:1, an enhanced novel nanocomposite with nanofibrillar structures including structured TPA-MO and PPy demonstrated improved charge capacity and an extremely precise capacitance of 870 F . Due to suitable microstructural characteristics, most notably definite surface area, that is significantly plagued by the molar fraction of reactants and post-synthesis calcination reaction temperatures [100], such a build-up, is linked to higher material tolerance. A composite electrode for supercapacitors was suggested using polyaniline electrochemically coated on a metallic element (tantalum) sheet as a matrix to place hydrous ruthenium (IV) oxide (RuO_2) particles. PANI provides a larger surface area for developing RuO_2 due to its porous nature. The particular capacitance (474 F/g), perfect recurring stability, and minor charge transfer resistance (2.24Ω) were found in the RuO_2 /polyaniline electrode [101]. Silver (Ag) has recently been mixed with PANI using a one-step electro-deposition process for the creation of an ultrahigh porosity composite suitable to be used as a pseudocapacitor electrode. After being evaluated in $0.5 \text{ M Na}_2\text{SO}_4$ at 10 mV/s , the silver/polyaniline composite revealed an ultrahigh specific capacitance of 420 F/g , which is nearly twofold greater than pure PANI. The better capacitive behaviour was attributed to higher electronic conductivity and express mass transit of chemical electrolytes ion (Na^+ and H^+) via the macropores of the Ag/PANI composite. Furthermore, however, after 2000 cycles, 94 percent of composites retained their novel capacitance [102]. A unique integrated composite contains polyaniline (PANI) components, anatase and rutile titanium dioxide (TiO_2) nanoparticles, and GO (graphene oxide) sheets as a future supercapacitor or diverse energy source system material. The electrode conductivity and consistency were enhanced by the addition of (TiO_2) nanoparticles and GO (graphene oxide). The PANI/ TiO_2 /GO composite had a large specific capacitance (1020 F/g at two mV/s and 430 F/g at one mV/s) and an extended cycle being (over thousand times) due to the near and cross-essential of the mechanisms, which fabricated a synergic implication and delivered apparent electron transport among the mechanisms [103]. One-dimensional nanostructured conducting polymers had been widely used as supercapacitor electrode materials.

Graphene/PANI hybrid electrode for supercapacitors is based on the electrostatic interactions between the negatively charged GO sheets and the positively charged emeraldine base form of PANI utilizing the layer-by-layer (LbL) technique [104].

To increase the mechanical permanence and electrical conductivity, the hybrid electrodes were reduced with hydrazine vapour and annealed after being constructed with stable suspensions of polyaniline (PANI) and graphene oxide (GO) in acidic conditions (Figure 10). While the discharge current density was 0.5 A/g , the gravimetric capacitance was 375.2 F/g . This layer-by-layer approach [104]

offers fine control over the thickness and internal structure of the graphene/polyaniline (PANI) hybrid supercapacitors and developed chemical stability and electronic conductivity during the charge-discharge method.

PANI nanofibres made utilizing a new approach that uses a fibre template path via a binary oxidant system are described as an electrode material for supercapacitors employing a quick periodic technique, with a large definite capacitance amount of 428 F/g. The permeable nanostructure provided not just a substantial fluid-dense interfacial surface and the prospect of electrolyte integration and exclusion, but furthers a small and simple route for ion transit [105]. Fabric materials with 3D exhibited nanowire design, such as polyaniline nanowire arranged electrodes created using an anodic confession process using a membrane-template synthesis approach, may be used in the form of supercapacitor electrode materials to give the large power density. The outstanding specific capacitance of 1142 F/g and high-rate capacity at a high current density of PANI nanowire arrayed electrodes inspired the development of a next-generation electrochemical supercapacitor. Because of the smaller diameter of the nanowires, PANI has a greater exploitation rate, resulting in exceptional capacitive abilities.

1.12. Batteries. Batteries have previously been described as energy storage devices. Stages of research for lightweight batteries include the use of novel materials such as conducting polymers and the expectation of increased discharge capacity. The polymers PANI [106], PPy [107], PTh [108, 109], PEDOT [110], and their composites [111] are favoured for use as electrodes in rechargeable batteries with lithium electrodes. When the first effective protest of chemically manufactured PANI residue as a cathode in desiccated (because of the low extent of electrolyte utilized) rechargeable batteries, the possibility of viable PANI rechargeable batteries was postulated. The addition of graphite and C_2H_2 black to PANI concentrate doped with perchlorate ion (ClO_4^-), as well as the subsequent mixing of mixture, improved cathode conductivity and therefore battery performance. Preferred compound powder applied to the negative electrode (combination of Zn powder, MgO, ZnO, and carboxymethyl cellulose (CMC) levelled to cylindrical nature) inhibits H_2 gas from progressing. During the first 100 charge-discharge sequences, the accumulated battery had an open-circuit voltage (OCV) of 1.64 V, a charge packing capacity of 125.43 mAh/g, and a columbic proficiency of more than 90% [112]. After recycled as cathode and anode material in a rechargeable battery containing Zn vessel as anode, cellulose acetate as the separator, and polyvinyl sulfate (PVSO₄) and carboxymethyl cellulose (CMC) as the solid polymer electrolyte (SPE), conducting polymer composites such as polyaniline (PANI/TiO₂) offer advanced charge-discharge appearances up to 50 cycles and better cyclability when compared to conducting polymer only [113]. SnO₂-Ppy nanocomposite formed using an one-pot in situ oxidative chemical polymerization technique also

demonstrated the improved cyclability and electrode dynamics for metallic elements such as lithium/tin (Li/Sn) alloying and de-alloying due to its 1D nanostructure and maintained tin oxide dispersion in polypyrrole matrix [114].

Composite electrode materials of advanced conductive polymers, such as polypyrrole (PPy), applied as thin layers on a proper enormous area apparent could be used to stimulate the act of nonmetal-based energy storage techniques. The development of a feasible nanostructured high-surface region electrode material for energy storage was started using cellulose fibres of algal origin that were exclusively layered with a 50 nm thin coating of polypyrrole (PPy). By combining traditional of metal-organic framework (MOF) materials (ZIF-67) and polypyrrole (PPy) through direct electrochemical deposition, Liu et al. [115] created a unique composite electrode material. Due to its mechanical flexibility and highly conductive textile properties, carbon cloth (CC) was special as a substrate. To fully utilize ZIF-67's edge in supercapacitors in this composite electrode, a "carbon-to-MOFs-to-polypyrrole" conducting network was created by growing PPy on the ZIF-67 surface. Electrochemical characteristics of MOFs were greatly enhanced through the electrochemical deposition of polypyrrole. As a result, the CC/ZIF-67/polypyrrole composite electrode displayed an exceptional capacitance of 180.7 mFcm² (284.3 F·g⁻¹). In addition, 100.7% ultrahigh long-life cycling stability was attained after 40,000 cycles. The reported MOF-based supercapacitor electrodes with this value were the best. The ZIF/PPy/CC composite, as a positive (+) electrode material, has the potential for supercapacitor applications due to its exceptional performance.

Consequences set the record for the highest claimed charge capacities and alleging rates for all conjugated polymer paper-based battery to date. Nanocomposite conductive paper material had an exact surface area of 80 m²·g⁻¹, and the batteries made from it were proficient in existence charged with currents as high as 600 mA/cm², with only a 6% capacity loss over 100 charge and discharge cycles [116]. Every effort has been made to advance the electrochemical performance of metallic element lithium (Li) ion batteries (LIBs). Several carbon materials with a wide range of micro/macrostructures are investigated. The electro-spinning was used to convert PPy polymer into continuous nanofibres (CNFs). Polyacrylonitrile (PAN)/30 wt% electrospun multiphase nanofibre PPy bicomponent precursors, which were previously employed as anodes to collect Li + half cells after removing any polymer binder or chemically conductive addition, shown high flexible capacity, increased cycle concert, and reasonable rate aptitude. Over 50 charge/discharge sessions, the outward fibrous shape was also maintained [117]. Innovative results impacting the valence functionalization of CNT with PVK, as well as additional applications of this composite in metallic element lithium (Li) rechargeable battery as a positive (+) electrode in conjunction with (lithium-hexafluorophosphate) LiPF₆ electrolytic solution, have been discussed. In lithium ion batteries, PVK functionalized single-walled carbon nanotubes (SWCNTs) and multi-walled carbon nanotubes (MWCNTs) had greater specific

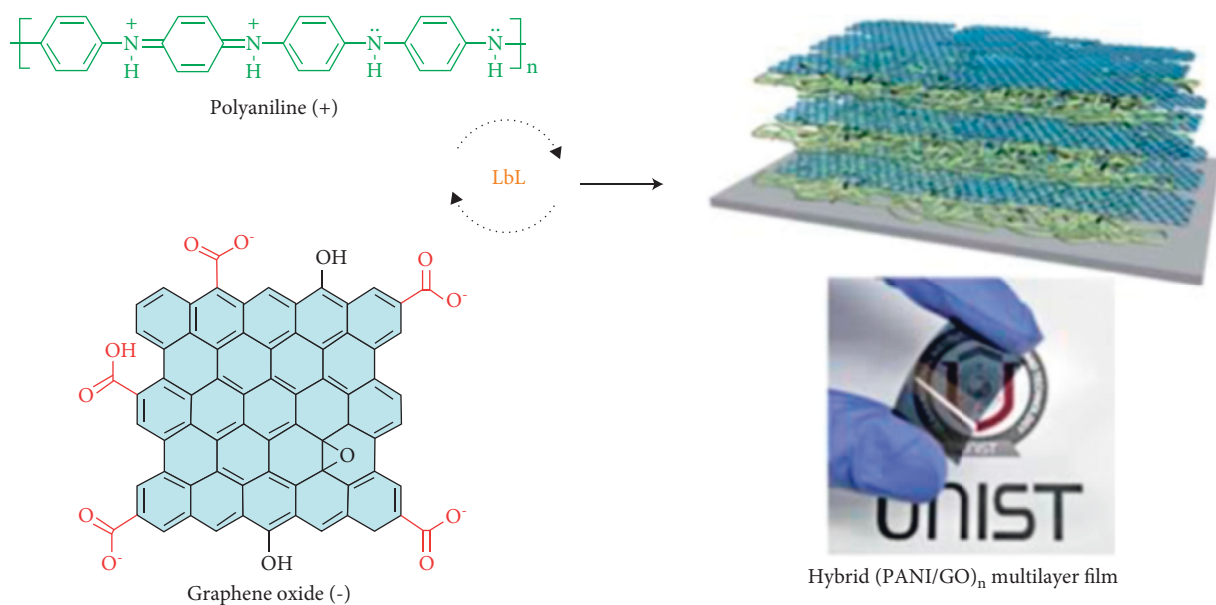


FIGURE 10: Polyaniline with graphene oxide using layer by layer of multilayer flexible film.

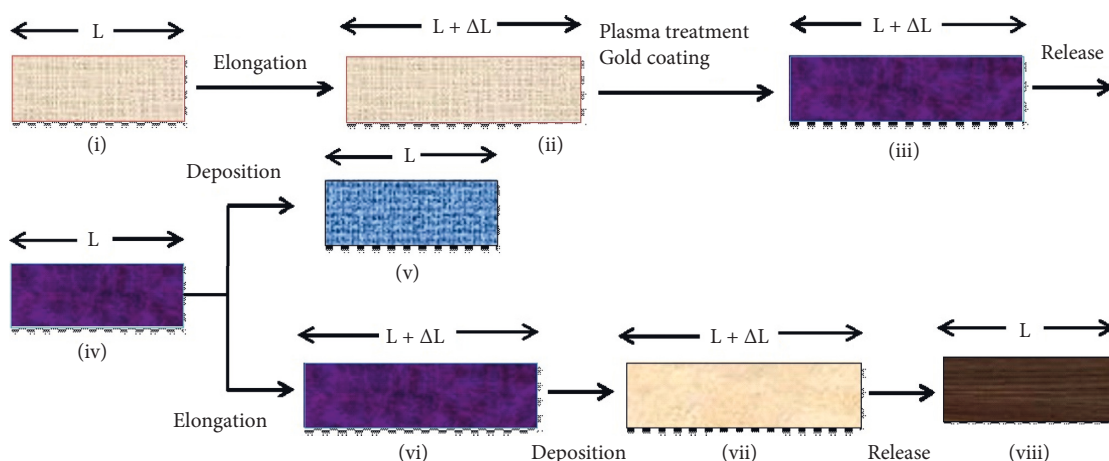


FIGURE 11: Diagram measures confirmed to organize buckled Au or polypyrrole PPy-pTS films on SIBS substrate of those products. Buckled PPy-pTS assembly involves the surface treatment, gold (Au) sputter coating, relaxation of the pre-strained SIBS substrate, and polypyrrole (PPy) electro-deposition. 2D buckled polypyrrole PPy-pTS film was electrodeposited on the re-elongated Au-layered substrate, achieved after the reduction in the substrate. (a) and (b) Buckled Au film (iv); (c) and (d) buckled polypyrrole (PPy)-pTS film (v); (e) and (f) 2D buckled polypyrrole (PPy) pTS film (viii). Inset, side outlook of PPy-pTS film showing the minor buckles.

discharge capacities of 45 and 115 mAh/g [118]. According to recent research, the shape of multiwalled carbon nanotubes (MWCNTs) to PANI generated via an ex situ and in situ oxidative chemical polymerization process has guaranteed the polymer, which is a noble electrochemical distinct as a cathode material in rechargeable lithium ion (Li) batteries. The inclusion of MWCNT results in a higher number of active sites for the faradic reaction, as well as a higher specific capacitance and enhanced electrical conductivity than pure PANI. As a result, the polyaniline/MWCNT composite has a better columbic efficiency, a larger discharge capacity, and the ability to sequence [119]. Wang et al. [120] have made a start towards integrating all polymer batteries into wearable and diverse textile

architectures. Over 40 cycles, researchers evaluated the concert of a fibre battery with polypyrrole/PF₆ cathode and polypyrrole/PSS anode specified capacity of 10 mAh/g. Using single-walled carbon nanotubes (SWCNTs) as anode material and polyvinylidene fluoride (PVDF) hollow fibre membrane as a separator, Wang was also able to develop the concert of the fibre battery. In their prior work on fibre batteries with a polypyrrole-hexafluorophosphate (PPy/PF₆) cathode and a polypyrrole (PPy)/PSS anode in the “flooded cells,” they revealed a twofold increase in capacity (about 20mAh/g). They found that when using polypyrrole-hexafluorophosphate (PPy/PF₆) as the cathode and single-walled carbon nanotubes (SWCNTs) as the anode in a “dry cell” rather

than a “flooded cell,” the capacity of the fibre battery was lower [121]. The electronic conductivity of the cathode can be improved by combining sulfur with conductive materials such as conducting polymers. Extended polysulfide lines were collective to PPy as side chains to source high definite capacity, and conducting polymer polypyrrole (PPy) was also used as consistent supports to create electric conducting networks. The sublimation S integration in tubular PPy structure was maintained by reducing surface area and pore volume with increasing sulfur concentration in the polypyrrole (PPy) composite. The improved cycling stability of the S/T-polypyrrole composite electrode was attributed to a mutual effect that increased conductivity and competitively diffused nano-sized S, as well as the formation of metallic element (Li) polysulfide between the T-polypyrrole matrix [122]. It was believed that the innovative synthetic technique, exchangeable energy via a simple one-step ball milling technique without heat treatment, would synthesize sulfur/polypyrrole dual composite and thus size up a high-energy lithium-sulfur (Li-S) secondary battery. Wang et al. [123] predicted a major flexible battery electrode based on buckling biocompatible polypyrrole-pTSA as the cathode material in magnesium (Mg) cells. The reduction of the pre-strained polypyrrole-pTSA deposited gold spit-coated poly(styrene-block-isobutylene-block-styrene) (SIBS) substrate illustrated in Figure 11 was important to the assembly operation. The material electrode withstood 2000 stretching cycles at 30% strain, gaining 1.2 V cell voltage [123].

2. Conclusions: Current Challenges and Prospects

The current state of conjugated polymeric materials is in the forms of synthesis processes, thermal, physical, chemical, morphological, optical, mechanical, and electronic feature engineering, and the evolution of modern and progressive energy storage and harvesting technologies. After passing through the rising phase and into the development phase, there are still plenty of opportunities for innovation; nonetheless, assets remain the primary guiding assumption. The important competitions for great throughput polymer-based manufacturing automation for viability and new research area are fully stated for success in the energy excellence phase. Our surroundings contain a significant amount of energy. Nature, too, has its own set of rules for converting solar, thermal, and vibration energy into electricity, the supreme entire energy form active in regular living. Only one part of the organic photovoltaic (PV) effect, the thermoelectric effect, has been investigated so far, and more work is needed to notice these effects to enhance the device structure and performance.

Conjugated polymer-based composites offer a lot of assistance in terms of electrical property development, chemical stability, and device concert without sacrificing the flexibility and processability of organic solar cell devices. In this article, we have conferred several organic photovoltaic (PV) technologies, sensor, organic solar cell

operation and supercapacitor, solar cell expressions and characterization parameters, conjugated polymer composites, and polymer-fullerene and polymer-inorganic hybrids, and DBA block polymer for appliances in OPVs. Furthermore, distinct device structures (conventional and inverted) were conferred on bilayer and BHJ solar cell designs. Aids and issues with device coordination and stability had also been discussed. The polymer/fullerene derivative BHJ composites, among the many conjugated polymer-based composites, are the most regarded in the development circumstances and also have achieved maximum cell efficiencies exceeding 10%. Other systems using a polymer-inorganic hybrid and a DBA block copolymer demonstrated higher charge transfer, but their efficiency was still significantly lower than that of the conjugated polymers fullerene composite-based gadgets, owing to inadequate charge transport or poor composite structure. One of the main competitions is to reduce polymer band gaps and cause HOMO/LUMO energy levels to grow best device performance. Though, the polymer synthesis process is extensively complicated. However, chemists are able to synthesise advanced novel polymers with low band-gap, enhanced HOMO/LUMO energy levels that improved match of the acceptor, and high hole mobility. Alternative significant competition is that single-junction organic polymer solar cells retain control over device efficiency. The future research will be focused on multijunction polymer organic solar cell topologies using the diversified composites.

Polymer materials are widely made at low cost using the solution or melting techniques, and they can be designed to be light and flexible. As a result, polymer materials are mostly well suited to stretchable and wearable electronics, a promising next-generation technology. For that purpose, more focus should be made on different ways to improve gadget efficiency by altering their structures. In particular, extending the stability of polymer-based devices is critical, as conducting polymer materials are known to be less stable than their inorganic equivalents. So far, energy harvesting systems have only appeared in a planar construction, and they have been designed to be thin to accommodate the flexible nature of current electronics. However, while these thin-film devices may be made into warped shapes, they are challenging to make three-dimensionally paradoxical.

In health monitors, biomedical, electronic sensors, flexible displays, and wearable, portable energy storage devices are just a few examples of the flexible and wearable electronic gadgets that have recently emerged as a prominent technical trend with explosive growth [124] batteries and flexible supercapacitors, for example, are in excessive demand. The most important factor in designing and making flexible and wearable supercapacitors is high safety, which is attended by noble mechanical stability in terms of stretching or twisting consistency, compression stability, and high energy density.

Recently, Xue et al. [124] introduced the different kinds and most current developments of electrode materials, such as carbon-based nanomaterials, inorganic metal and metal oxide materials, and advanced conductive polymers. Despite the

advances made in flexible and wearable supercapacitors so far, there are still certain issues. For instance, it is still difficult to prepare flexible electrodes with strong mechanical characteristics and outstanding electrochemical stability, etc. As a result, generating highly conductive electrodes with a high energy density is quite appealing. To achieve various purposes, more solid/gel electrolytes must possess strong ionic conductivity, chemical stability, and good mechanical performance. Conjugated polymers are found suitable for flexible energy harvesting and storage devices [125–128].

A twisting deformation, on the other hand, is required for a wide range of flexible electronic goods. In this regard, a novel family of 1D (one-dimensional) fibre-shaped energy collecting devices is strategically important and has attracted rising benefits in recent times. These energy harvesting fibres might also be perverted into a variety of shapes with no visible structural consequences. Furthermore, they may be woven into more flexible electronic textiles and are capable of wearable applications, which are a rapidly growing field, but the energy conversion efficiency of such as fibre-shaped harvesting energy devices is small, and supplying them at a large scale is problematic. The poor electrical conductivity of the utilized fibre electrode has been confirmed as the most significant constraint on the rather directions. In the future, a significant amount of work should be put into developing higher conducting fibres and fabrication conducts for fibre devices.

Data Availability

The data presented in this study can be obtained from the corresponding author upon request.

Conflicts of Interest

The authors declare that there are no conflicts of interest regarding the publication of this study.

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